# Cambridge (CIE) A Level Maths: Pure 1

Your notes

## Integration

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- Fundamental Theorem of Calculus
- Integrating Powers of x
- Definite Integration
- Reverse Chain Rule
- Improper Integrals



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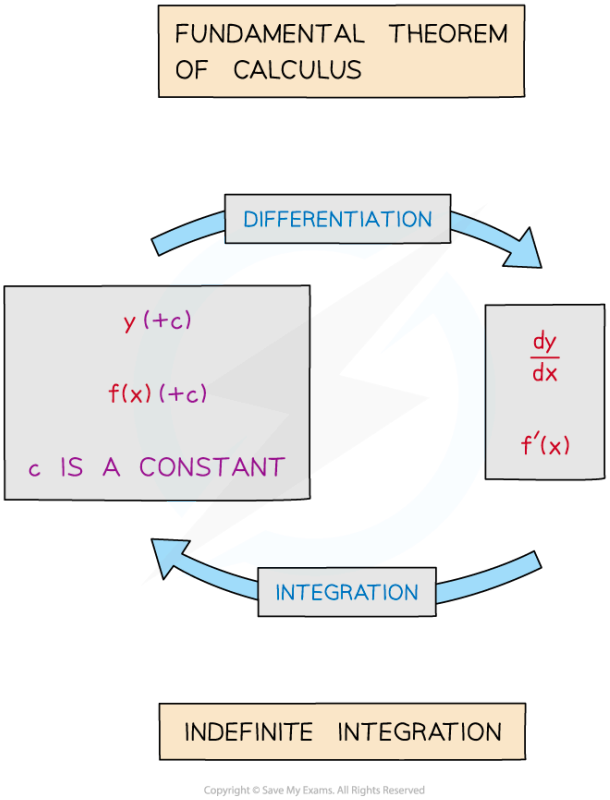
Your notes

# Fundamental theorem of calculus

## What is the fundamental theorem of calculus?




- The Fundamental Theorem of Calculus states that integration is the inverse process of differentiation
  - This form of the Theorem relates to **Indefinite Integration**
  - An alternative version of the Fundamental Theorem of Calculus involves **Definite Integration**

## What is a constant of integration?

- When differentiating **y**, constant terms 'disappear'
  - for constants **y = c**, $\frac{dy}{dx} = 0$
  - graphs of constants are horizontal lines and so have gradient $\left(\frac{dy}{dx}\right)$ of 0

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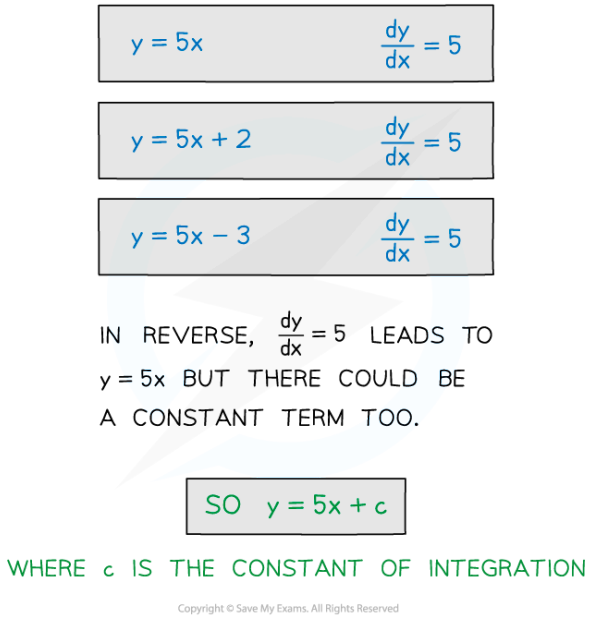
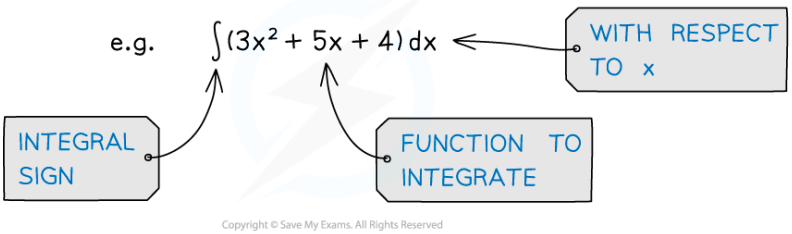
- Integrating $\frac{dy}{dx}$, to get **y**, cannot determine the constant
  - To acknowledge this constant, "**+ c**" is used
  - **c** is called the **constant of integration**




## What notation is used in integration?




- $\int$ is the sign for integration
- If it has more than one term the function to be integrated (called the **integrand**) should be in brackets
  - "Integrate" -- "**all** of (...)" -- "with respect to x"
- **dx** means integrate with respect to **x**, any other letter is treated like a number (ie like a constant)

## Worked Example

Your notes

(a) Find $\frac{dy}{dx}$ for $y = 6x + 3$

(b) Find $\int 6 \, dx$

a) $\frac{dy}{dx} = 6$

b) $y = 6x + c$

CONSTANT OF INTEGRATION

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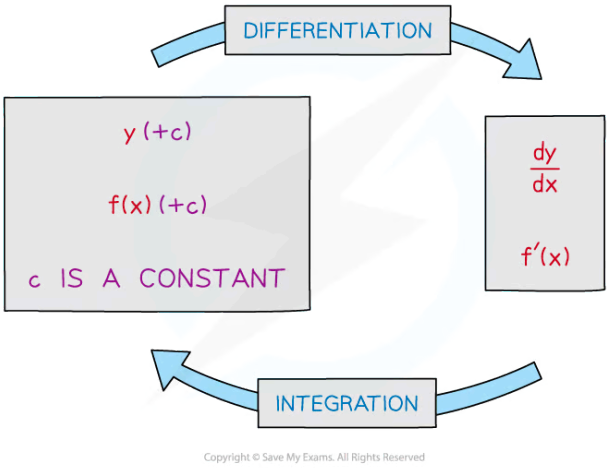
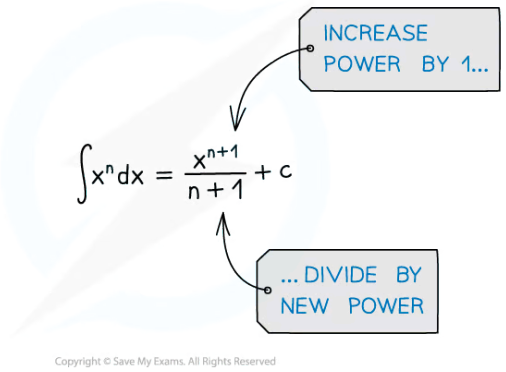
# Integrating powers of x

## What is integration?

- Integration is the **inverse** ('opposite') of differentiation




## How do I integrate powers of x?




- For each term ...
  - ... increase the power (of x) by 1
  - ... divide by the new power
- This does not apply when the original power is -1
  - the new power would be 0 and division by 0 is undefined

# Finding the constant of integration

## How do I find the constant of integration?

Your notes

- STEP 1 Rewrite the function into a more easily **integrable** form
  - Each term needs to be a power of **x** (or a constant)
- STEP 2 Integrate each term and remember "+c"
  - Increase power by 1 and divide by new power
- STEP 3 Substitute the coordinates of a given point in to form an equation in **c**
  - Solve the equation to find **c**

e.g. FIND THE EQUATION OF THE GRAPH THAT PASSES THROUGH (2,6) AND HAS GRADIENT FUNCTION $3x^2(2x-1)$

ANOTHER NAME FOR $\frac{dy}{dx}$

STEP 1 REWRITE IN A MORE EASILY INTEGRABLE FORM

$$\frac{dy}{dx} = 3x^2(2x-1)$$

$$= 6x^3 - 3x^2$$

STEP 2 INTEGRATE...

$$y = \frac{6x^4}{4} - \frac{3x^3}{3} + c$$

... REMEMBERING "+c"

$$y = \frac{3x^4}{2} - x^3 + c$$

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STEP 3 USE THE GIVEN POINT TO FIND c

AT (2,6) $6 = \frac{3}{2}(2)^4 - 2^3 + c$

$c = -10$

$\therefore \quad y = \frac{3}{2}x^4 - x^3 - 10$

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## Worked Example

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Given $f'(x) = \dfrac{(x+3)^2}{\sqrt{x}}$ and $f(1) = 25$,

find $f(x)$.

STEP 1

$$f'(x) = \frac{x^2 + 6x + 9}{x^{\frac{1}{2}}}$$

REWRITE IN A MORE EASILY INTEGRABLE FORM

$$f'(x) = \frac{x^2}{x^{\frac{1}{2}}} + \frac{6x}{x^{\frac{1}{2}}} + \frac{9}{x^{\frac{1}{2}}}$$

$$f'(x) = x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + 9x^{-\frac{1}{2}}$$

STEP 2

$$f(x) = \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{9x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

INTEGRATE EACH TERM

REMEMBER "+c"!

$$f(x) = \frac{2}{5}x^{\frac{5}{2}} + 4x^{\frac{3}{2}} + 18x^{\frac{1}{2}} + c$$

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STEP 3

$$f(1) = \frac{2}{5} + 4 + 18 + c = 25$$

FIND c

$$c = \frac{13}{5}$$

$$\therefore\ f(x) = \frac{2}{5}x^{\frac{5}{2}} + 4x^{\frac{3}{2}} + 18x^{\frac{1}{2}} + \frac{13}{5}$$

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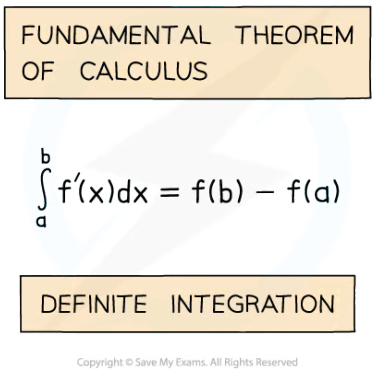
# Definite integration

## What is definite integration?

- **Definite Integration** occurs in an alternative version of the Fundamental Theorem of Calculus
- This version of the Theorem is the one referred to by most AS/A level textbooks/websites




- **a** and **b** are called limits
  - **a** is the lower limit
  - **b** is the upper limit
- **f'(x)** is the **derivative** of **f(x)**

## Why is there no constant of integration in definite integration?

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e.g. FIND $\int_2^5 9x^2\,dx$

$f'(x) = 9x^2$

$\therefore\ f(x) = 3x^3 + c$

INDEFINITE INTEGRATION

$f(2) = 3(2)^3 + c = 24 + c$

$f(5) = 3(5)^3 + c = 375 + c$

$\therefore\ f(5) - f(2) = (375 + c) - (24 + c)$

$= 375 - 24 + c - c$

$= 351$

DEFINITE INTEGRATION

c's CANCEL

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- "**+c**" would appear in both **f(a)** and **f(b)**
  - Since we then calculate **f(b)** – **f(a)** they cancel each other out
  - There would be a "**+c**" from **f(b)** and a **–**"**+c**" from **f(a)**
- So "**+c**" is **not included** with definite integration

## How do I find the value of a definite integral?

- STEP 1: If not given a name, call the integral
  - This saves you having to rewrite the whole integral every time!
- STEP 2: If necessary rewrite the integral into a more easily integrable form
  - Not all functions can be integrated directly
- STEP 3: Integrate without applying the limits
  - Notation: use square brackets [ ] with limits placed after the end bracket
- STEP 4: Substitute the limits into the function and calculate the answer

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e.g. EVALUATE $\int_1^9 (2x^2 + \frac{3}{x^2} - 4\sqrt{x})\,dx$

STEP 1 NAME THE INTEGRAL, I

LET $I = \int_1^9 (2x^2 + \frac{3}{x^2} - 4\sqrt{x})\,dx$

STEP 2 REWRITE IN A MORE EASILY INTEGRABLE FORM

$I = \int_1^9 (2x^2 + 3x^{-2} - 4x^{\frac{1}{2}})\,dx$

STEP 3 INTEGRATE

- USE SQUARE BRACKETS
- LIMITS AT END

$I = \left[\frac{2}{3}x^3 - 3x^{-1} - \frac{8}{3}x^{\frac{3}{2}}\right]_1^9$

STEP 4 SUBSTITUTE LIMITS AND CALCULATE

$I = \left(486 - \frac{1}{3} - 72\right) - \left(\frac{2}{3} - 3 - \frac{8}{3}\right)$

$I = \frac{1256}{3}$

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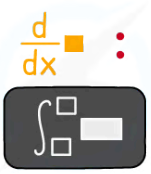
## How do I find a definite integral on my calculator?

- Advanced scientific calculators can work out the values of definite integrals
- The button will look similar to:



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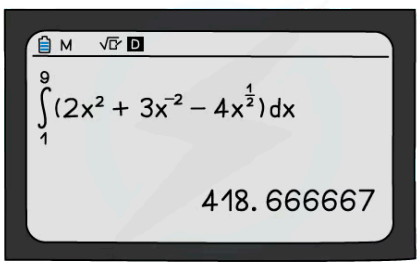


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- (Note how the calculator did not return the exact value $\left(\frac{1256}{3}\right)$ of the integral)

## Examiner Tips and Tricks

- Look out for questions that ask you to find an **indefinite** integral in one part (so "**+c**" needed), then in a later part use the same integral as a **definite** integral (where "**+c**" is not needed).

## Worked Example

Find the value of

$$\int_2^4 3x(x^2 - 2)\,dx$$

Start by expanding the brackets inside the integral

$$\int_2^4 (3x^3 - 6x)\,dx$$

Integrate as usual (here it's a 'powers of $x$' integration)

Write the answer in square brackets with the integration limits outside

$$\int_2^4 (3x^3 - 6x)\,dx = \left[3\left(\frac{x^{3+1}}{3+1}\right) - 6\left(\frac{x^{1+1}}{1+1}\right)\right]_2^4$$

$$= \left[\frac{3}{4}x^4 - 3x^2\right]_2^4$$

Now substitute 4 into that function
And subtract from it the function with 2 substituted in

$$\left[\frac{3}{4}x^4 - 3x^2\right]_2^4 = \left(\frac{3}{4}(4)^4 - 3(4)^2\right) - \left(\frac{3}{4}(2)^4 - 3(2^2)\right)$$
$$= (192 - 48) - (12 - 12)$$
$$= 144 - 0$$
$$= 144$$

$$\int_2^4 3x(x^2 - 2)\ dx = 144$$

Your notes


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# Reverse Chain Rule

## How do you integrate $(ax + b)^n$?

- The reverse chain rule can be used for integrating functions in the form $y = (ax + b)^n$
  - Make sure you are confident using **the chain rule** to **differentiate** functions in the form $y = (ax + b)^n$
  - The reverse chain rule works backwards
- For $n = 2$ you will most likely **expand the brackets** and **integrate each term separately**
- If $n > 2$ this becomes **time-consuming** and if $n$ is not a positive integer we need a different method completely
- To use the **reverse chain rule** $\int (ax + b)^n \, dx$ (provided $n$ is not -1)
  - **Raise** the **power of $n$ by 1**
  - **Divide** by this **new power**
  - **Divide** this whole function by the **coefficient of $x$**
    - $$\int (ax + b)^n \, dx = \frac{(ax + b)^{n+1}}{n+1} \times \frac{1}{a} + c$$
- You can **check your answer by differentiating it**
  - You should get the original function when you differentiate your answer
- Note that this method **only works** when the function in the brackets is **linear ($ax + b$)**

### Worked Example

Your notes

Evaluate

$$\int_1^2 \frac{1}{(\sqrt{3x-1})}\,dx$$

giving your answer as an exact value.

$$\int_1^2 \frac{1}{\sqrt{3x-1}}\,dx = \int_1^2 (3x-1)^{-\frac{1}{2}}\,dx$$

INCREASE THE POWER BY 1

$$= \left[\frac{(3x-1)^{\frac{1}{2}}}{\frac{1}{2}} \times \frac{1}{3}\right]_1^2$$

DIVIDE BY THE NEW POWER

DIVIDE THE WHOLE FUNCTION BY THE COEFFICIENT OF THE X INSIDE THE BRACKET

$$= \left[\frac{2\sqrt{3x-1}}{3}\right]_1^2$$

$$= \frac{2\sqrt{3(2)-1} - 2\sqrt{3(1)-1}}{3}$$

$$= \frac{2\sqrt{5} - 2\sqrt{2}}{3}$$

$$\int_1^2 \frac{1}{\sqrt{3x-1}}\,dx = \frac{2\sqrt{5} - 2\sqrt{2}}{3}$$

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## Examiner Tips and Tricks

Make sure you can recognise when a question needs to use the reverse chain rule, it may not always be obvious.

Your notes

# Improper Integrals

## What are improper integrals?

- An **improper integral** is a definite integral where one or both of the limits is either:
  - Positive or minus infinity
  - A point where the function is undefined
    - Consider the graph of $y = \frac{1}{\sqrt{x}}$
    - It is undefined at the point $x = 0$
    - The integral of $y = \frac{1}{\sqrt{x}}$ with a **limit of zero** would be an improper integral
- Examples include:
  - $\int_0^5 \frac{1}{\sqrt[3]{x}} \, dx$
  - $\int_1^\infty \frac{1}{x^3} \, dx$


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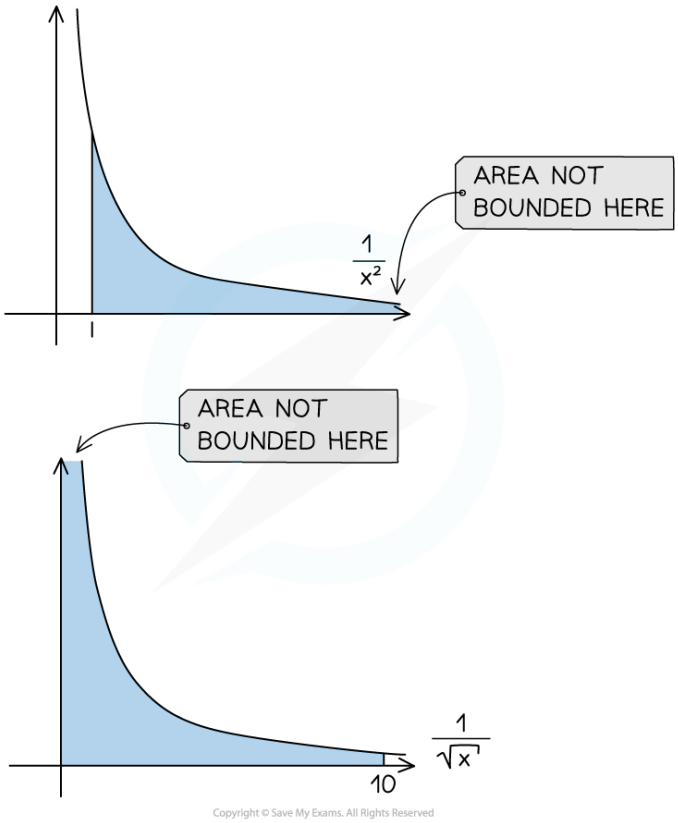


## How do we find the value of an improper integral?

- Use algebra to **replace the limit** which cannot be found with a variable
  - E.g. let the undefined limit of zero be *a* or the infinite limit be *b*
- **Evaluate the integral** and substitute your chosen variable into the expression
- Consider what will happen to your answer as the value of your chosen variable **tends towards the limit**
  - E.g. what happens as *a* gets closer to zero or as *b* gets closer to infinity?
- Your final answer will be the value you get if you substitute this into your answer
  - E.g. as *a* tends to zero $a^2$ tends to zero and so this part of your solution will be zero
- It is useful to remember as *a* tends to infinity then $\frac{1}{a}$ tends to 0

### Worked Example

Your notes

Find the following improper integrals, give your answers as exact values.

(a) $\int_0^3 \frac{1}{2\sqrt{x}}\,dx$

(b) $\int_3^\infty \frac{1}{2x^2}\,dx$

a) $\int_0^3 \frac{1}{2\sqrt{x}}\,dx$

UNDEFINED AT X = 0

$$\int_a^3 \frac{1}{2\sqrt{x}}\,dx = \left[\frac{1}{2}x^{\frac{1}{2}}\right]_a^3$$

USE a AS THE LIMIT INSTEAD OF 0

$$= \left[\frac{\frac{1}{2}x^{\frac{1}{2}}}{\frac{1}{2}}\right]_a^3$$

$$= \left[x^{\frac{1}{2}}\right]_a^3$$

$$= \sqrt{3} - \sqrt{a}$$

CONSIDER HOW $\sqrt{a}$ CHANGES AS $a \to 0$

AS $a \to 0$ $\sqrt{a} \to 0$ so $\sqrt{3} - \sqrt{a} \to \sqrt{3}$

$$\int_0^3 \frac{1}{2\sqrt{x}}\,dx = \sqrt{3}$$

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b) $\int_3^\infty \frac{1}{2x^2}\,dx$

LIMIT OF INFINITY CANNOT BE FOUND

USE b AS THE LIMIT

$$\int_3^b \frac{1}{2x^2}\,dx = \left[\frac{1}{2}x^{-2}\right]_3^b$$

$$= \left[\frac{\frac{1}{2}x^{-1}}{-1}\right]_3^b$$

$$= \left[\frac{-1}{2x}\right]_3^b = -\frac{1}{2b} - \left(-\frac{1}{6}\right) = \frac{1}{6} - \frac{1}{2b}$$

CONSIDER HOW $\frac{1}{2b}$ CHANGES AS $b \to \infty$

AS $b \to \infty$ $\frac{1}{2b} \to 0$ SO $\frac{1}{6} - \frac{1}{2b} \to \frac{1}{6}$

$$\int_3^\infty \frac{1}{2x^2}\,dx = \frac{1}{6}$$

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## Examiner Tips and Tricks

- Be careful if a limit of your integral is zero, always check to see if the function is defined at zero and if not treat it as an improper integral.

- Infinite limits will always be treated as improper integrals.

Your notes